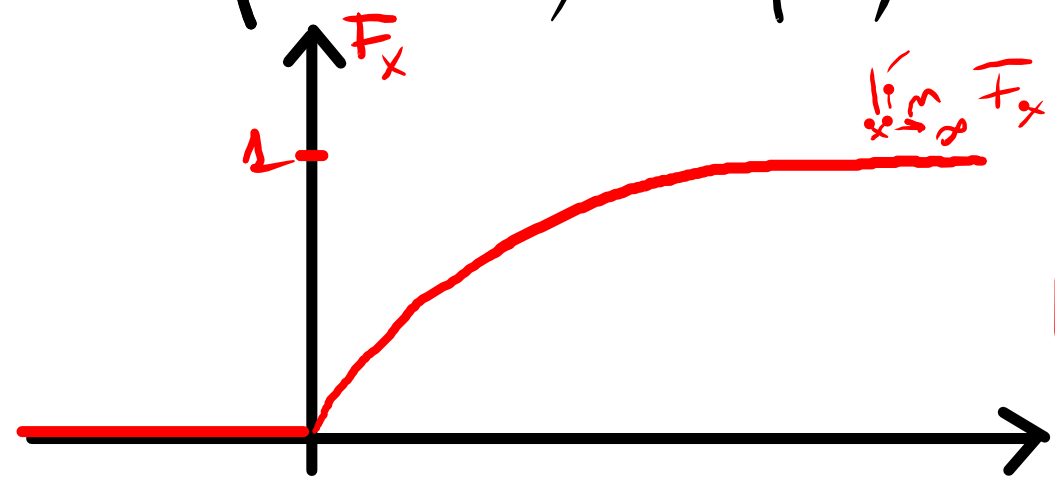
EXPONENCIAL: $X \sim Exp(\lambda)$ $\lambda > 0$

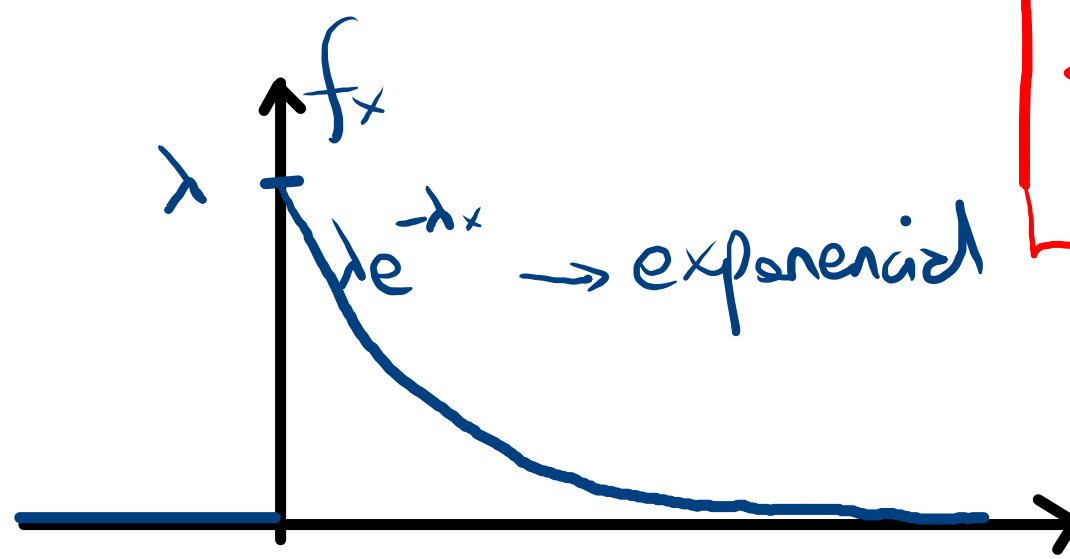
↳ tiempos de vida, de espera, entre sucesos...

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-\lambda x} & x \geq 0 \end{cases}$$



$\lim_{x \to \infty} F_X(x) = 1$

$$f_X(x) = \begin{cases} 0 & x < 0 \\ \lambda e^{-\lambda x} & x > 0 \end{cases}$$



$$\boxed{\begin{aligned} \mu_X &= \frac{1}{\lambda} \\ \sigma_X^2 &= \frac{1}{\lambda^2} \therefore \sigma_X = \frac{1}{\lambda} \end{aligned}}$$

Resultado conocido: $\int_0^\infty t^{n-1} e^{-t}\, dt = (n-1)!$

$$\int_0^\infty \lambda e^{-\lambda t}\, dt = \int_0^\infty e^{-u}\, du = 0! = 1 \checkmark$$

$n=1$

$\boxed{\begin{array}{l} u = \lambda t \\ du = \lambda\, dt \end{array}}$

$n=2$

$$E[X] = \int_0^\infty t\, \lambda e^{-\lambda t}\, dt = \int_0^\infty \frac{u}{\lambda} e^{-u}\, du = \frac{1!}{\lambda} = \frac{1}{\lambda}$$

$n=3$

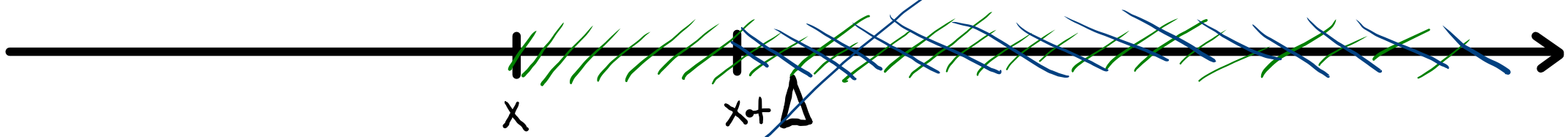
$$E[X^2] = \int_0^\infty t^2 \lambda e^{-\lambda t}\, dt = \int_0^\infty \frac{u^2}{\lambda^2} e^{-u}\, du = \frac{2!}{\lambda^2} = \frac{2}{\lambda^2}$$

$$Var(X) = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

$$\boxed{\begin{array}{l} \mu_X = \frac{1}{\lambda} \\ \sigma_X^2 = \frac{1}{\lambda^2} \\ \sigma_X = \frac{1}{\lambda} \end{array}}$$

$X \sim Exp(\lambda)$ → modelan tiempos

$$P(X > x) = 1 - F_X(x) = 1 - \left(1 - e^{-\lambda x}\right) = e^{-\lambda x} \qquad \forall\, x > 0$$



$$P(X > x+\Delta \mid X > x) = \frac{P(X > x+\Delta,\ X > x)}{P(X > x)} = \frac{P(X > x+\Delta)}{P(X > x)} =$$

$$= \frac{e^{-\lambda(\cancel{x}+\Delta)}}{e^{-\lambda \cancel{x}}} = e^{-\lambda \Delta} = P(X > \Delta)$$

FALTA DE MEMORIA

Ejemplo: X = tiempo de vida de un dispositivo

$$P(X > 20+1 \mid X > 20) = P(X > 1)$$